

Cover Letter — The Flex (Software Engineer)

Headline — *LinkedIn note to the hiring team / email opener:*

I've already built a short-term rental platform from scratch — and did the market analysis behind it. Python/FastAPI, AWS, PostgreSQL, distributed APIs, and AI built into the product: RAG property matching and multimodal search by image, location or description.

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Dear Hiring Team,

I'm applying for the Software Engineer role. The combination you're hiring for — Python and FastAPI, AWS, PostgreSQL, API-driven architecture, automation and AI integration, on a proptech platform — is unusually close to work I've already done, in two places.

The proptech part. I joined a friend's company, Trastain, as technical partner on an equity stake and built a short-term rental booking platform from scratch. I ran the competitor analysis across Airbnb-style platforms first, concluded that the Airbnb model isn't culturally aligned with the Egyptian and wider Middle Eastern market, and turned that into the product's positioning. On the engineering side: shared-property booking, preference matching, policy enforcement, transaction security, Stripe and Paymob integration with webhook and HMAC verification — and **RAG-based property matching plus multimodal search**, so users can query by image, by location, or by natural-language description.

The platform part. As an R&D engineer at Andalusia I build Python services on FastAPI — Pydantic v2, async/await, SQLAlchemy 2.0 over PostgreSQL, tested with pytest — inside a distributed architecture with Temporal Saga orchestration, RabbitMQ messaging, service discovery and an API gateway. I also build the automation layer: an AI-driven pipeline that handles requirements analysis, story generation, test creation, Playwright test automation, and pull-request review that comments line-by-line and checks code against its business requirements. That's five custom MCP servers and a code graph across 38 modules. "Systems that create leverage at scale" is a fair description of it.

Cloud. I've worked across both providers and chosen between them deliberately — fully managed Azure services where speed to MVP mattered, self-managed **AWS EC2 and S3** where control did — and cut messaging costs by moving off a cloud SMS service to a local provider.

One honest gap: my front end is TypeScript and JavaScript rather than production **React**. I can read and reason about it and I'd get productive quickly, but I won't claim it as experience I don't have. My strength is clearly backend, APIs and AI integration.

I'm based in Alexandria, so the remote-first setup works well.

Best regards, Moamen Basyoni